Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/20/2012
PI Name:	Bedford, Joel Ph.D.		
Project Title:	Dose-Rate Effects and Components of Systems Governing Variations in Susceptibility for Carcinogenic and Acute Radiation Risks following Gamma-Ray, Proton, or HZE Irradiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	970-491-7492
Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 5/31/2012 (previously 11/30/2011), per NSSC information (Ed., 11/29/2011) NOTE: End date is now 11/30/2011, per NSSC information (Ed., 5/31/2011) NOTE: Received NCE through 5/31/2011, per C. Guidry/JSC (08/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nagasawa , Hatsumi (Colorado State University) Peng, Yuanlin (Colorado State University) Weil, Michael (Colorado State University)		
Grant/Contract No.:	NNX07AP85G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The central hypothesis is that dose-rate effects in the 1 to 0.005 Gy/min range will be the same for protons vs. gamma-rays for chromosomal aberration induction. Second, levels of chromosomal radiosensitivity, and their dose-rate dependence will be similar for cells from individuals with genetic differences in radiosensitivity. Such a biomarker may help identify differences among individuals in a component contributing to their susceptibility to these effects. The aims for the year reported below corresponding to FY2009 were: 1) to compare dose-rate effects for gamma-rays vs. protons for chromosomal aberration induction in cultured human fibroblast and epithelial cells. (This would test assumptions regarding equivalence of gamma-ray and high energy protons with respect to RBEs and dose-rate effects of interest for astronauts.) 2) to determine whether changes in the 2D culture or 3D tissue-like context of cells may affect the utility of the putative markers for predicting carcinogenic and acute effects radiosensitivity. For low passage normal human fibroblasts and for MCF10A human mammary epithelial cells, the dose rate effect for induction of chromosomal aberrations seen for gamma-rays in the range of 70cGy/min (HDR) to 0.005cGy/min (LDR) was virtually identical for protons with energies of 1000MeV or 150MeV. For the MCF10A cells the responses (aberration induction, cell killing and gamma H2AX formation at low dose rates) in 2D or 3D culture were identical.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of variation among individuals in their susceptibility to radiation carcinogenesis and the genetic determination of the extent of existing variation would be important information, not only of interest to individual astronauts who will be exposed to radiation during space missions, but would also be of interest to anyone who may be exposed to radiation in their occupation or in connection with activities or experiences where radiation exposure may be involved.
Task Progress:	The major aims of this research centers around answering a very practical question on dose-rate effects for protons from solar particle events (SPE) that might be experienced by astronauts, as well as dose-rate effects from HZE radiations, such as 1GeV/n Fe ions. The question is being addressed using chromosomal aberrations in low passage cultured normal human cells as a surrogate system. This would help indicate the possible necessity for carrying out more complicated animal carcinogenesis proton or HZE ion dose-rate experiments. Details of the results and analyses are now being carried out and publications are being prepared. [Ed. note: compiled from FY2011 technical report, submitted October 2011, per PI's aproval 4/20/2012]
Bibliography Type:	Description: (Last Updated: 02/04/2013)
Articles in Peer-reviewed Journals	Nagasawa H, Brogan JR, Peng Y, Little JB, Bedford JS. "Some unsolved problems and unresolved issues in radiation cytogenetics: a review and new data on roles of homologous recombination and non-homologous end joining." Mutat Res. 2010 Aug 14;701(1):12-22. Review. PubMed PMID: 20298803 , Aug-2010